

Central African Republic Time Series Analysis

STA 137
Final Project

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Introduction

The economy of the Central African Republic is largely supported by agriculture, which contributes roughly half of the country's GDP. This sector accounts for about 80% of the total labor force. The Central African Republic contains an abundance of natural resources and has a low level of manufacturing power as it mainly focuses on exporting raw materials to bring in income, specifically in diamonds, timber, cotton, and coffee. Overall, the country is sparsely populated and rural and has an extremely high poverty rate. This makes both industrializing and having a strong labor force challenging. That being said, exports may or may not have a significant impact on their economy as measured by the GDP.

Analyzing the country's available GDP, export, and population data from the years of 1960 to 2017 will help stakeholders better understand how these variables change over time and interact with each other. Time series data analysis will allow us to see certain inflection points and times of strain on the economy and the population, and their effects on other metrics. Additionally, this study will uncover any underlying trends of each variable for further future inspection.

Exploratory Data Analysis

Based on our initial plots in Figure 2, our immediate observations were that none of the categories were stationary since they each exhibited an increasing or decreasing time trend. This indicated the necessity to difference the data. Trends over time for population and GDP climb upwards, while there is a clear downward trend for exports. Additionally, from an initial observation of the plots in Figure 2, it does not look like there are outliers. From our summary statistics, we realized that the mean and median are very closely aligned for all variables except for population, meaning they may not be normally distributed. The mean is higher than the median in the population, showing that there have been high points of population growth that bring the average population size up. Although there may not be any visually evident outliers, as per the norm of economic analysis, we may want to consider a log transform on the data to make the numbers easier to work with while simultaneously taming any effect from missed outliers.

Model Selection

GDP

We first examined the GDP variable in the data set. As discussed in our initial exploration, in order to stabilize possible variance and fix heteroskedasticity, we applied a log transformation. After completing the log transformation, we did not think it was necessary to do the Box-Cox as the log transformation was satisfactory.

The exploratory data analysis (Figure 2) showed that the data were non-stationary. As a result, we differentiated the data (Figure 4) to make the data stationary. The PACF plots for the

non-differenced data demonstrated a rapid decline after lag 1, while tailing off for the ACF. This suggested the use of an AR model since they generally follow this behavior. We initially chose ARIMA(1,1,0) after differencing because the sharp decline at lag 1 indicated that this was all that was necessary. Regardless, we considered ARIMA(1,1,0), ARIMA(0,1,1), and ARIMA(1,1,1) models as they were all still plausible.

The following are our models: **ARIMA(1,1,0): $\Delta y_t = 0.1642 \Delta y_{t-1} + \varepsilon_t$ | ARIMA(0,1,1): $\Delta y_t = \varepsilon_t + 0.1267 \varepsilon_t$ | ARIMA(1,1,1): $\Delta y_t = 0.978521 \Delta y_{t-1} + \varepsilon_t - 0.913256 \varepsilon_{t-1}$ |** We ultimately selected ARMA(1,1,1) as our final model due to it having the lowest BIC and AIC values, which indicate the best predictive power. Additionally, every coefficient had a p-value below our significance level of 0.05. Our standardized residuals charts (Figure 5) again supported our selection of this model. The Ljung box plot p-values are all above the significance threshold, further suggesting that the model's residuals appeared as white noise. The variance of the residuals is uniform, and there do not seem to be many outliers except for on the tails of the QQ plot. These outliers are acceptable for time series modeling as they only exist on the tails. Overall, the residuals behave like white noise, showing that this model has a good fit.

Using the fitted ARIMA(1,1,1) model, we forecasted for three years for both the transformed and the raw data (Figure 6). The forecast showed a mostly upward trend in GDP. When comparing the model with data from Worldometer on the Central African Republic GDP, the model aligned with the values presented in GDP for 2018, 2019, and 2020. This further supported that our model had a good fit.

Overall, our analysis demonstrated a strong fit for an ARMA(1,1,1) model and predicted a strong upward forecast in GDP. In addition, based on residual diagnostics, along with AIC and BIC values, we confirmed that the model has a good fit. The rising GDP may be attributed to the improvements in key economic sectors like agriculture and forestry. The strong performance of the ARMA(1,1,1) model demonstrates that it has lots of practical applications, including supporting economic planning in the Central African Republic, informing policy decisions, and serving as a short-term economic forecast when other data is unavailable.

Exports

Similar to GDP, the export data also required a log transformation. The original export data exhibited a substantial amount of variance (Figure 2), which indicated heteroskedasticity. Applying a log transformation helped to normalize the variance, reducing the influence of outliers.

After applying the log transformation, first-order differencing was applied to the data. This is due to the negative export growth time trend (Figure 2). Additionally, we saw that ACF plots saw a similar slow decay as in GDP (Figure 7), and PACF again showed a significant dip after lag 2. This again suggested to us an AR model as it follows this behavior. We also initially chose ARIMA(2,1,0) after differencing because the sharp decline at lag 2 indicated that this was all

that was necessary. Still, we considered the plausibility of ARIMA(2,1,0), ARIMA(0,1,1), and ARIMA(2,1,1) models.

These are the fitted models. **ARIMA(2,1,0): $\Delta y_t = -0.2974 \Delta y_{t-1} + \varepsilon_t$ | ARIMA(0,1,1): $\Delta y_t = \varepsilon_t - 0.3473 \varepsilon_{t-1}$ | ARIMA(2,1,1): $\Delta y_t = -0.0511 \Delta y_t + \varepsilon_t - 0.3094 \varepsilon_{t-1}$** | We selected ARIMA(2,1,0) as our final model since it had the lowest BIC and AIC values, which indicate the best predictive power. Also, every coefficient had a p-value below our significance level of 0.05. Based on the residual plots (Figure 9), the model is significant. There is uniform and constant variance across the standardized residuals plots. The QQ plot shows only minor deviations at the tail, and the majority of Box test p-values lie above the significance threshold. Altogether, the residual diagnostics show that residuals behave like white noise, supporting our findings that this model has a good fit.

In our forecast of our ARIMA(2,1,0) model with first-order differencing (Figure 10) for the years 2018, 2019, and 2020, both our transformed model and our raw population predict a decrease in exports. When comparing our graph to the Central African Republic export data from the World Bank, our model forecast matches the decrease in exports seen in the country in those three years.

The export data required log transformation and first-order differencing to improve heteroskedasticity and stationarity, which ultimately led us to choose an ARIMA(2,1,0) model. The model showed a strong fit with low AIC and BIC values compared to other models and clean standardized residual plots. The forecast also matched the real export data decline in growth. The decline in growth may be due to a decrease in the amount of natural resources found in the Central African Republic over time, which is why exports have declined over time. Our results show that this model could be used in many practical applications, whether for trade analysis or economic policy.

Population

The population data in the Central African Republic exhibits little variance; however, growth is not consistent, and increases over time. This is due to population growth being inherently not linear. For this reason, we applied a log transformation to the data to help linearize growth, stabilize any and all variance, and reduce the impact from any shocks or outliers in the data.

We observed a time trend in the population data, with increasing population growth (Figure 2), which informed us that we needed to perform first-order differencing. We saw that ACF plots had a similar slow decay as in GDP/Exports (Figure 11), and PACF again showed a significant dip after lag 1. Since this indicated an AR model, we initially chose ARIMA(1,1,0). Once again, we investigated multiple models.

These are the fitted models. **ARIMA(1,1,0): $\Delta y_t = 0.9954 \Delta y_{t-1} + \varepsilon_t$ | ARIMA(0,1,1): $\Delta y_t = \varepsilon_t + 0.9999 \varepsilon_{t-1}$ | ARIMA(1,1,1): $\Delta y_t = 0.0181 + 0.9568 \Delta y_{t-1} + \varepsilon_t + 0.9999 \varepsilon_{t-1}$** | We selected ARIMA(1,1,0) as our final model since it had the lowest BIC and AIC values, which indicate the

best predictive power. Also, every coefficient had a p-value below our significance level of 0.05, so we were confident in this model. However, after performing diagnostic checks on this model, we found that white noise still existed. In the Ljung box analysis (Figure 13), all the p-values lie below the significance threshold. However, the residual variance was constant, and the tails of the QQ plot for the residuals were acceptable. Due to all p-values lying below the significance threshold, we knew that we may have to difference again because of the possible implication of residuals not behaving like white noise. After plotting the ACF of the differenced model (Figure 12), we saw that the graph still showed a slow decay, indicating non-stationarity.

These are the newly fitted models. **ARIMA(1,2,0): $\Delta^2 y_t = 0.8932 \Delta^2 y_{t-1} + \varepsilon_t$ | ARIMA(0,2,1): $\Delta^2 y_t = \varepsilon_t + 0.9994 \varepsilon_{t-1}$ | ARIMA(1,2,1): $\Delta^2 y_t = 0.8534 \Delta^2 y_{t-1} + \varepsilon_t + 0.8062 \varepsilon_{t-1}$ |** We finally selected the model ARMA(1,2,1). The standardized residuals (Figure 14) showed a constant and uniform variance. We also saw that in the QQ plot, most of the data lies on the reference line with little deviation at the tails. Lastly, the Ljung-Box test showed strong proof of model fit, as all p-values lie above the significance threshold. This confirmed that the residuals now behaved like white noise and that the model had a good fit.

Lastly, we forecasted the data (Figure 15), which demonstrated an increase in population over the three years following 2017. Our forecast is consistent with data from Worldometer, which also shows an increase in population growth from 2018-2020 in the Central African Republic.

Overall, the second-order difference ARMA(1,2,1) model provided a strong fit for the population data. The residuals analysis confirmed the white noise behavior of the residuals, and the transformation of the data removed any exponential growth and extreme values in the data. The forecast closely matched our predicted values of increased population growth over three years. Our model demonstrates a stable long-term growth pattern in the population, and may be useful for population planning, forecasting future resource needs, and informing long-term policy decisions.

Cross-Correlation Function

Lastly, we wanted to determine the cross-correlation between GDP and Exports after removing autocorrelation. In Figure 16, the cross-correlation function plot of residuals between the final GDP and final exports models demonstrates a negative correlation at lag 0, meaning exports and GDP move together in opposite directions over time. However, at other lags, there is no large spike, which indicates that one cannot be used to predict the other's future, as they would both be impacted equally at the same time. Despite this, the plot further emphasizes their strong contemporary ties and effect on each other at the moment.

Conclusion

Based on our forecasts from each model, it appears that in the near future of this country, omitting any unexpected exogenous shocks, the Central African Republic's GDP and population are expected to rise, while its exports are expected to slowly drop. This aligns with our cross-correlation discovery that the country's GDP and exports move in opposite directions. In

the short term, our models provide solid forecasts with strong predictive power that can be used to determine the near-future trend. This can be of use to policymakers or other interested parties who may wish to understand where the country is headed economically. The most interesting and valuable finding in our analysis is still the relationship between GDP and exports over time. This connection implies that exports will become a smaller share of the GDP over time, possibly hinting toward a conclusion that another domestic sector, such as agriculture, may be more influential in the future on economic growth. It may be especially important for policymakers as they may be able to leverage this discovery by, for example, putting more resources into these other sectors instead, as a result multiplying the economic effect of this sectoral shift. Ultimately, these forecasts have a plethora of utility, and we only hope that the right party finds this information useful.

Appendix

Summary Statistics

Variable	Mean	Median	SD	Min	Max
GDP	939447947	934787385	615865973	112155599	2195599557
Growth	1.2	2.05	6.33	-36.7	9.48
CPI	73.58	71.42	30.44	36.77	186.86
Imports	30.83	29.68	7.13	18	46.9
Exports	20.66	21.74	5.94	10.68	34.31
Population	2974789.53	2842456	1052735.9	1503508	4659080

Figure 1

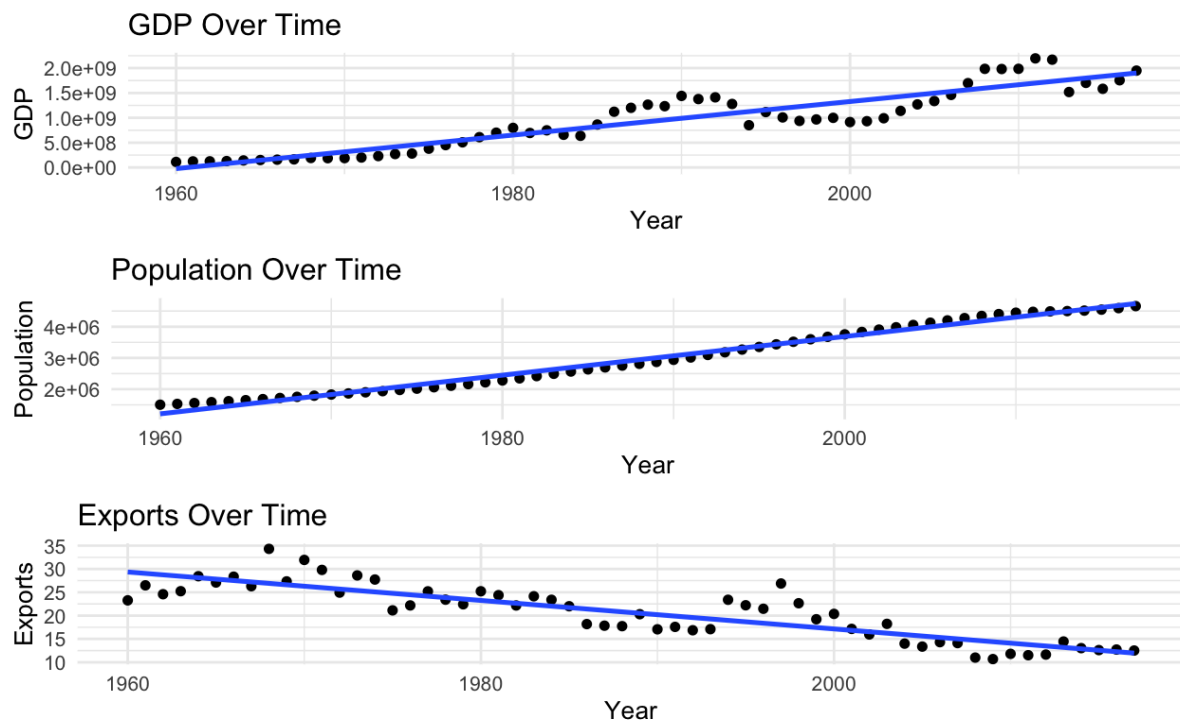


Figure 2

Economic Data Graphs

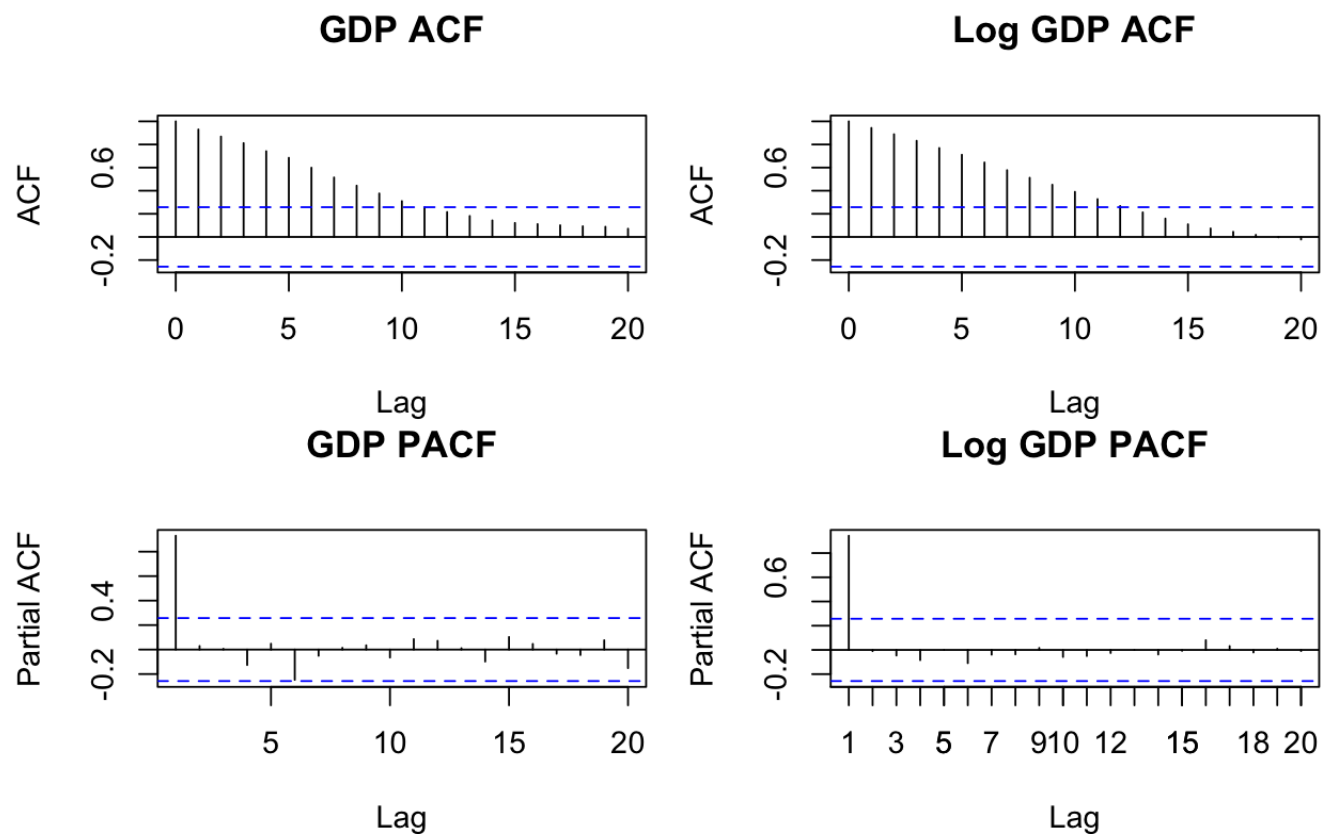


Figure 3

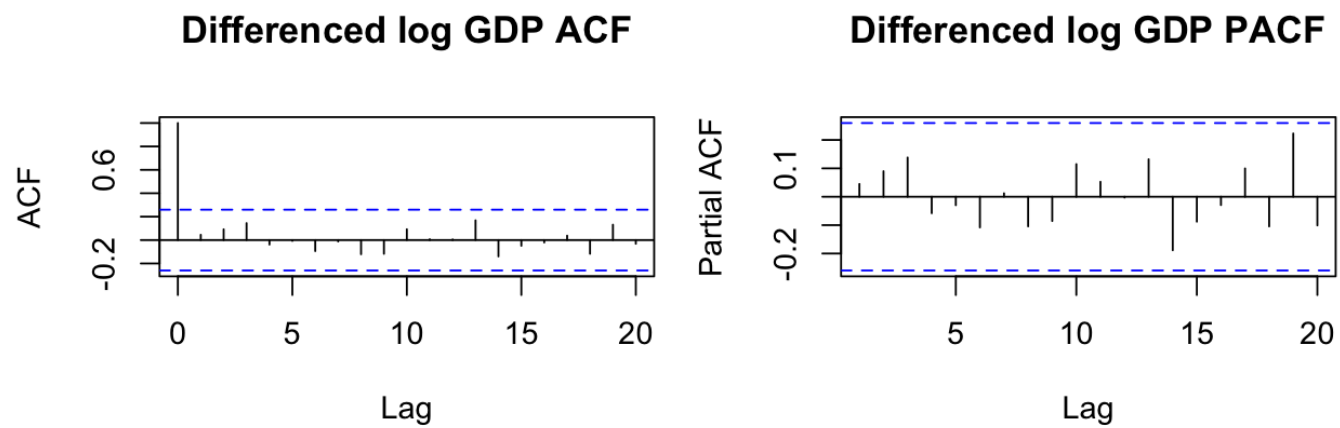


Figure 4

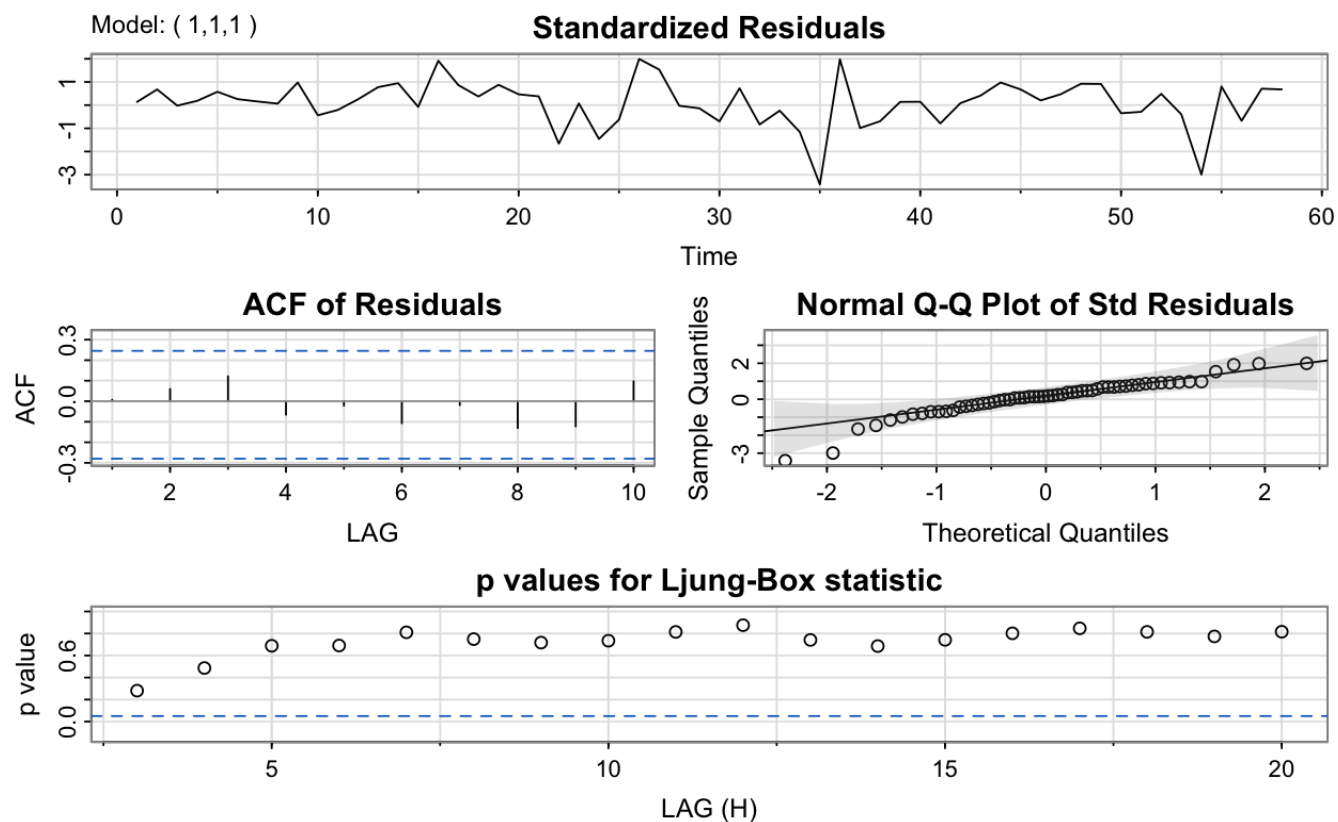


Figure 5

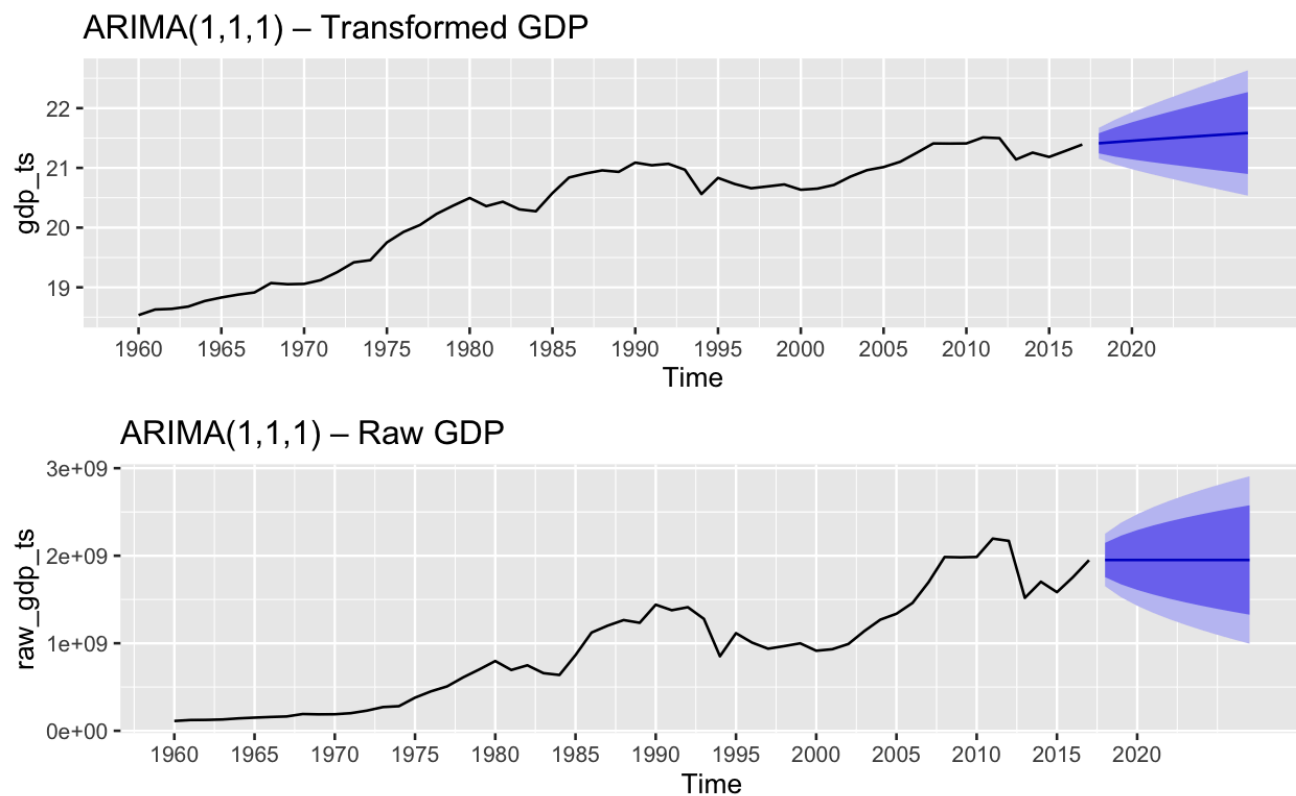


Figure 6

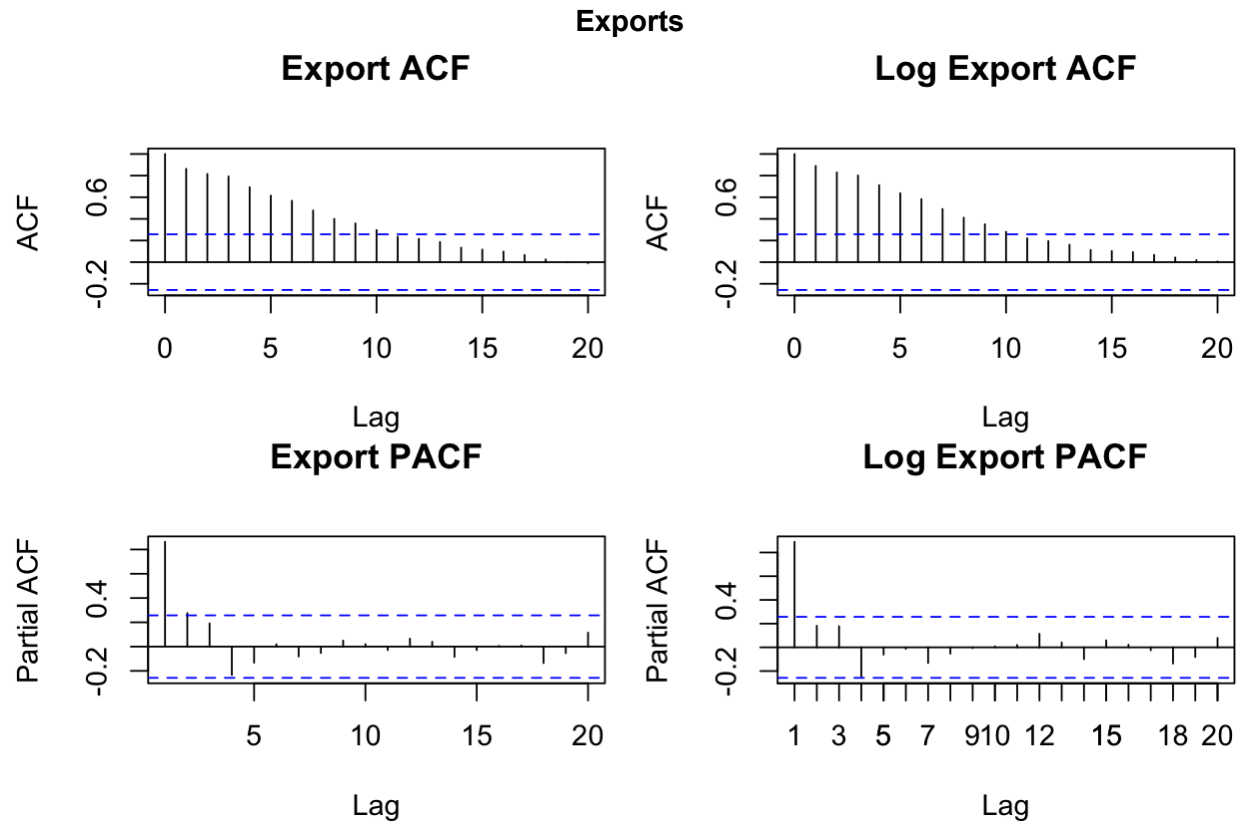


Figure 7

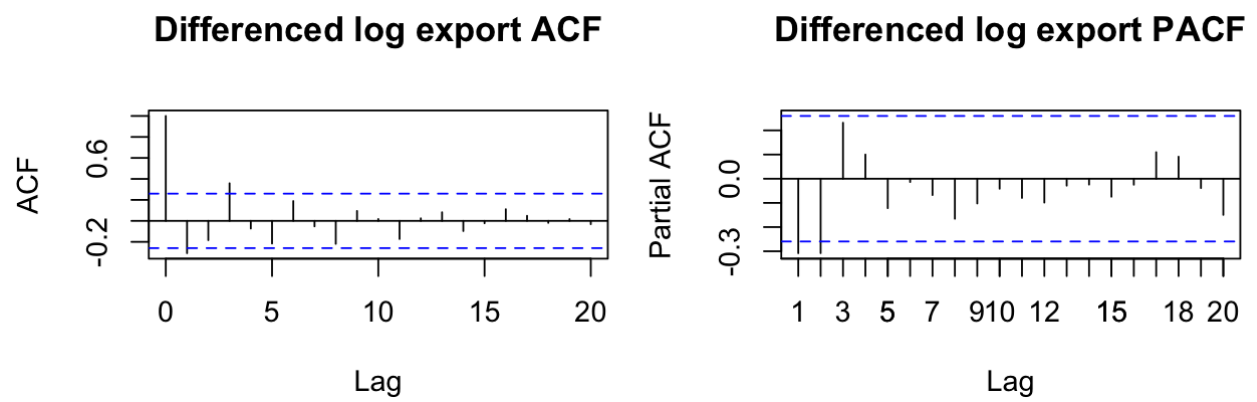


Figure 8

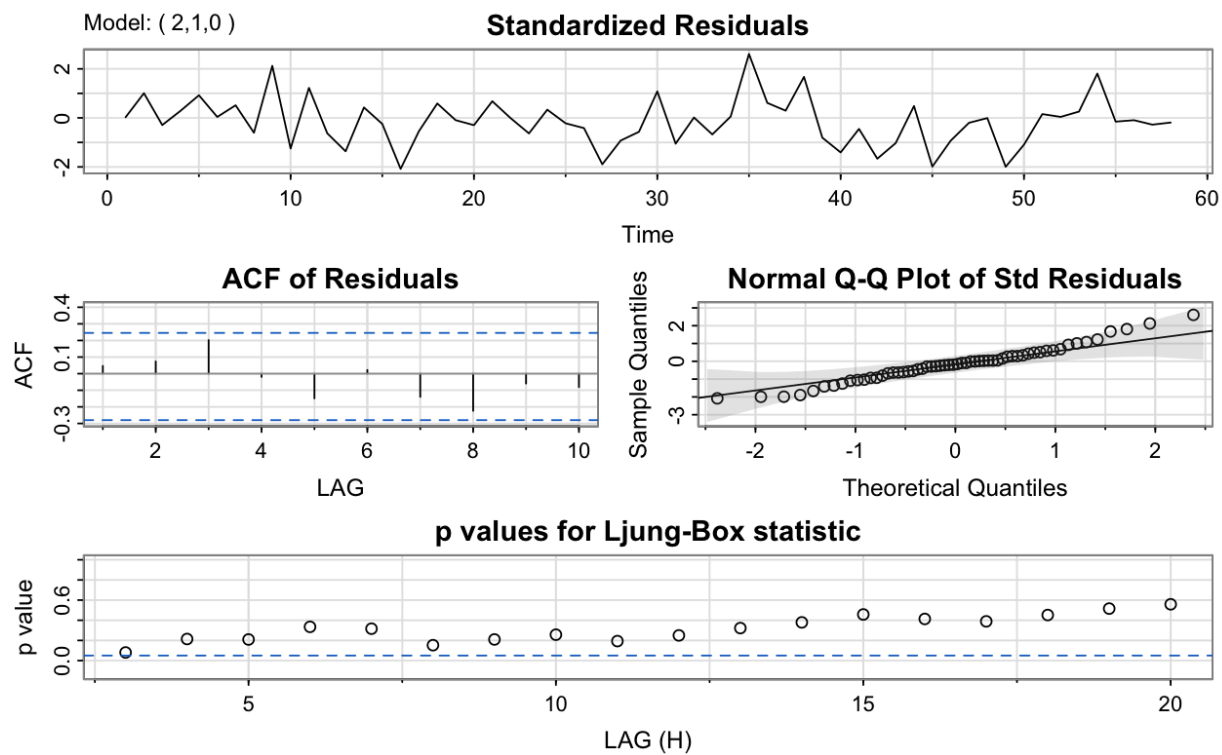


Figure 9

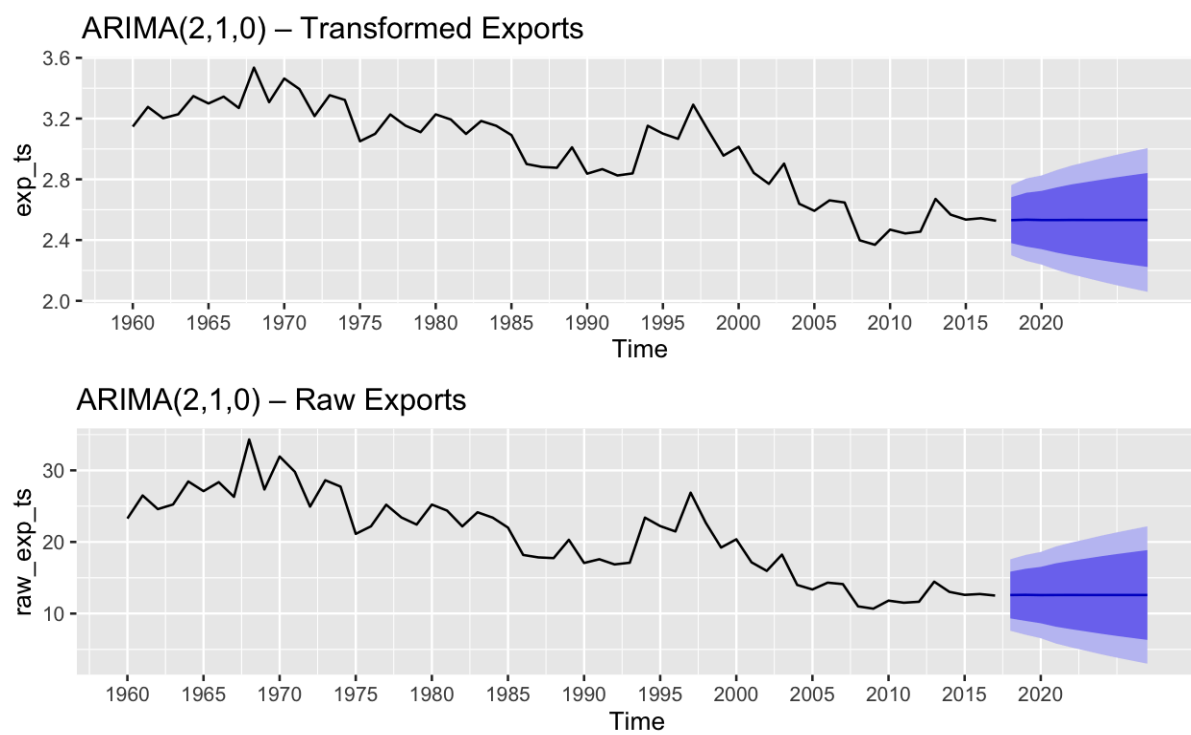


Figure 10

Population

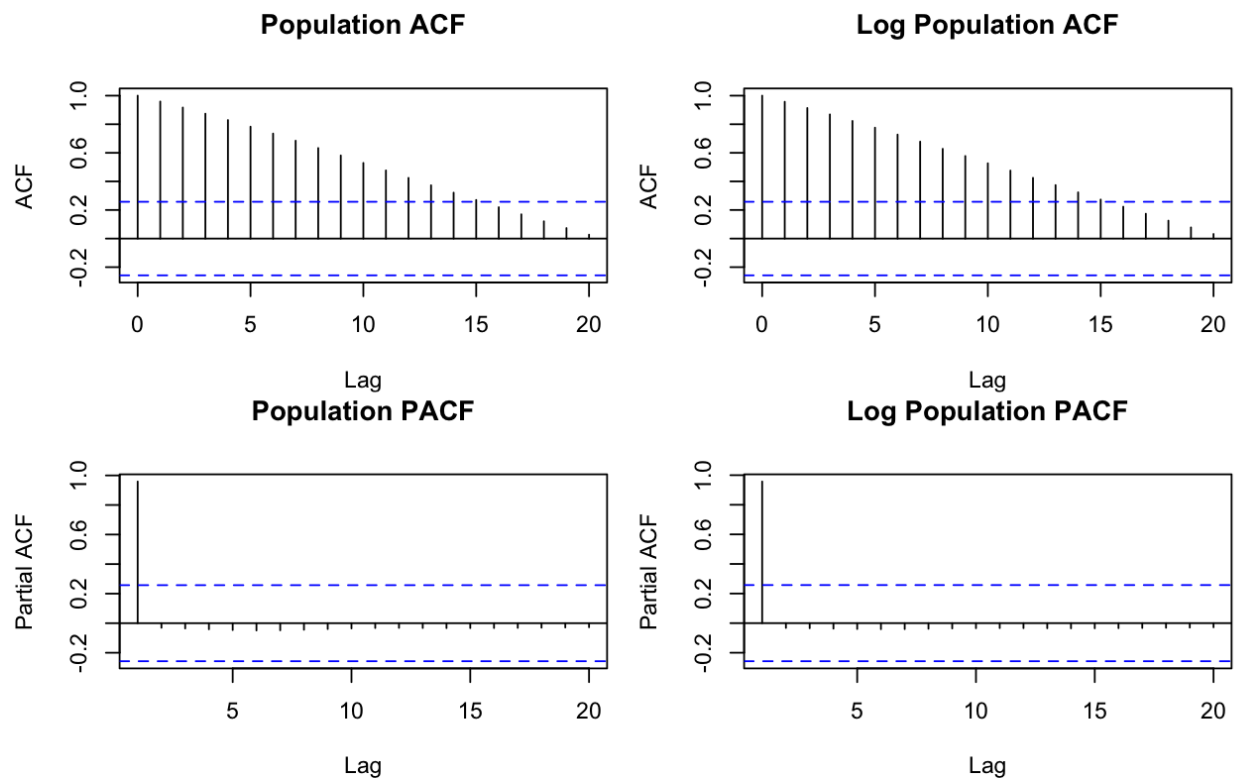


Figure 11

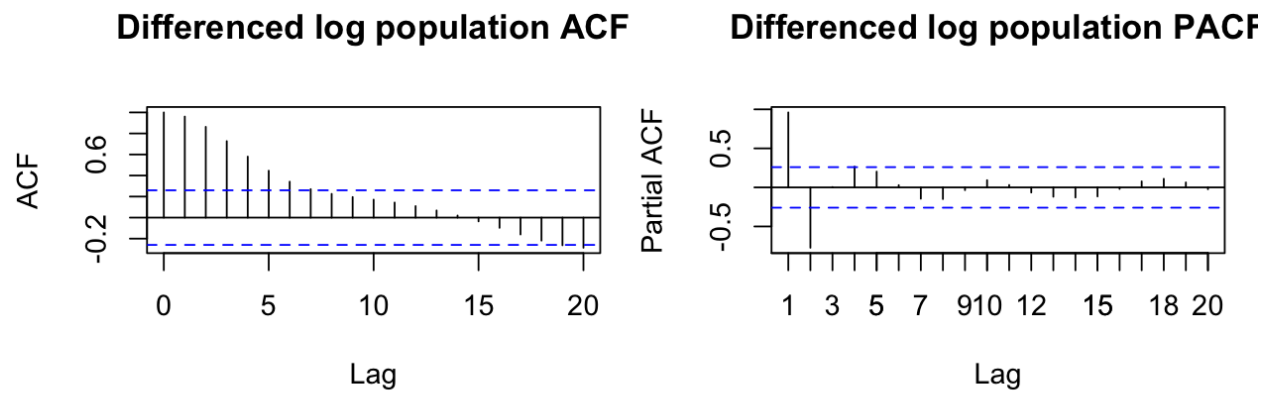


Figure 12

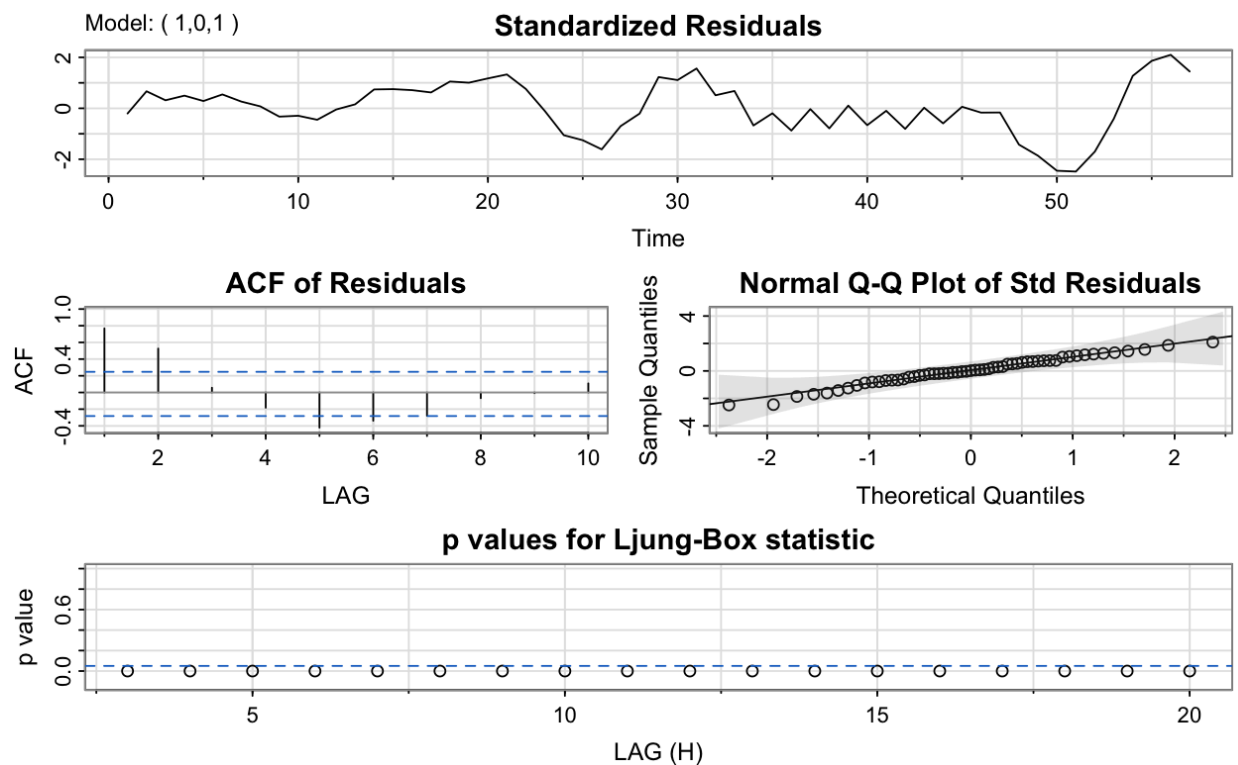


Figure 13

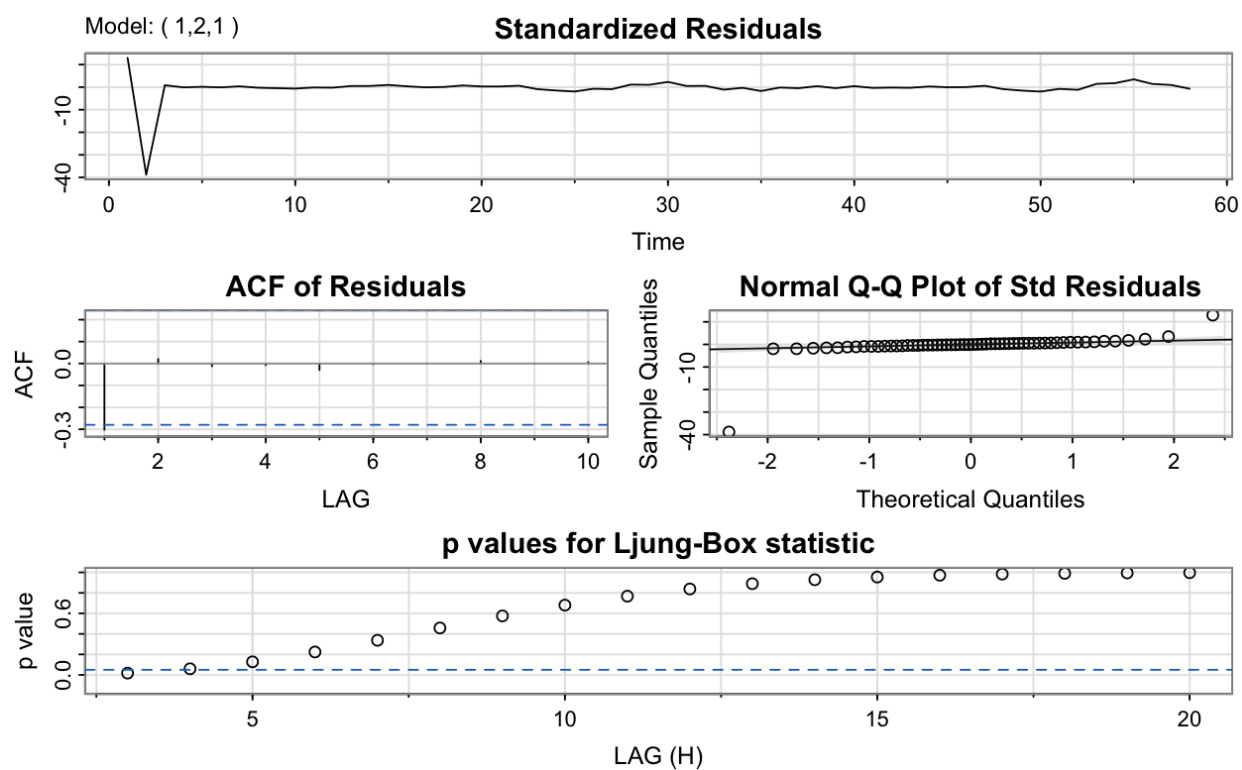


Figure 14

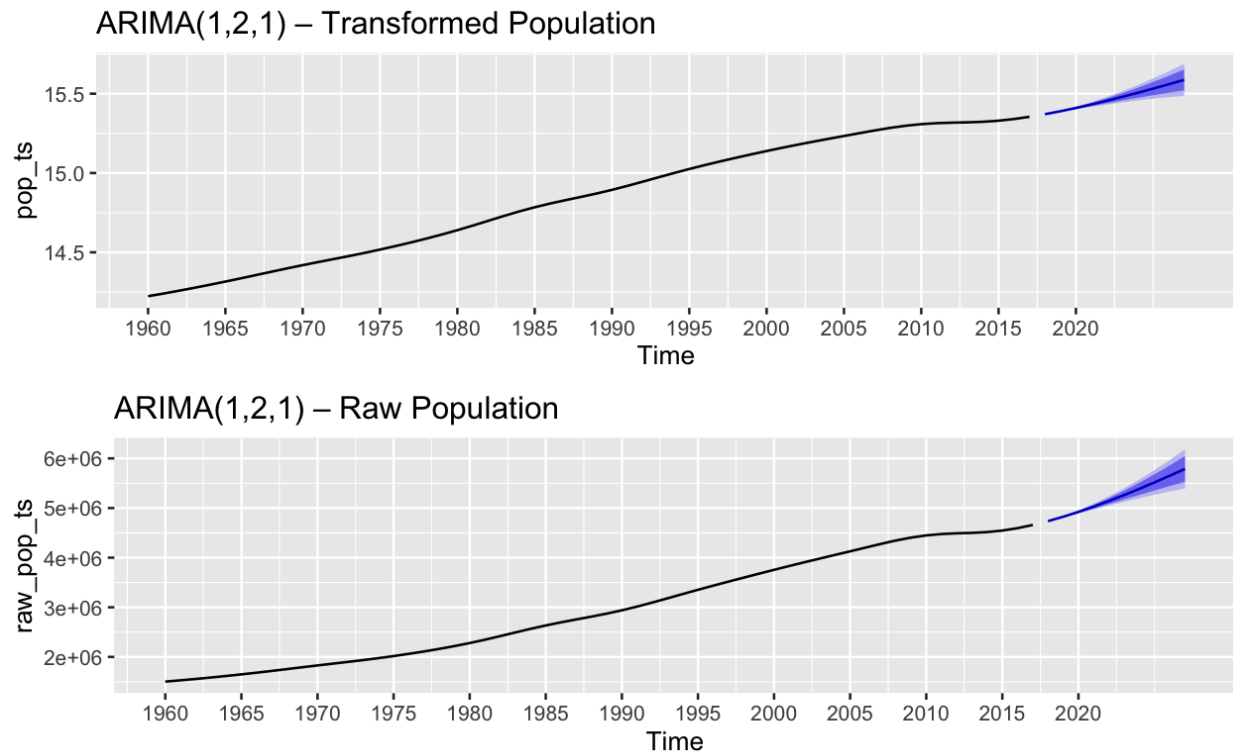


Figure 15

CCF

Export(ARMA(1,1,1) and GDP(ARMA(2,1,0)) Cross Correlation Function

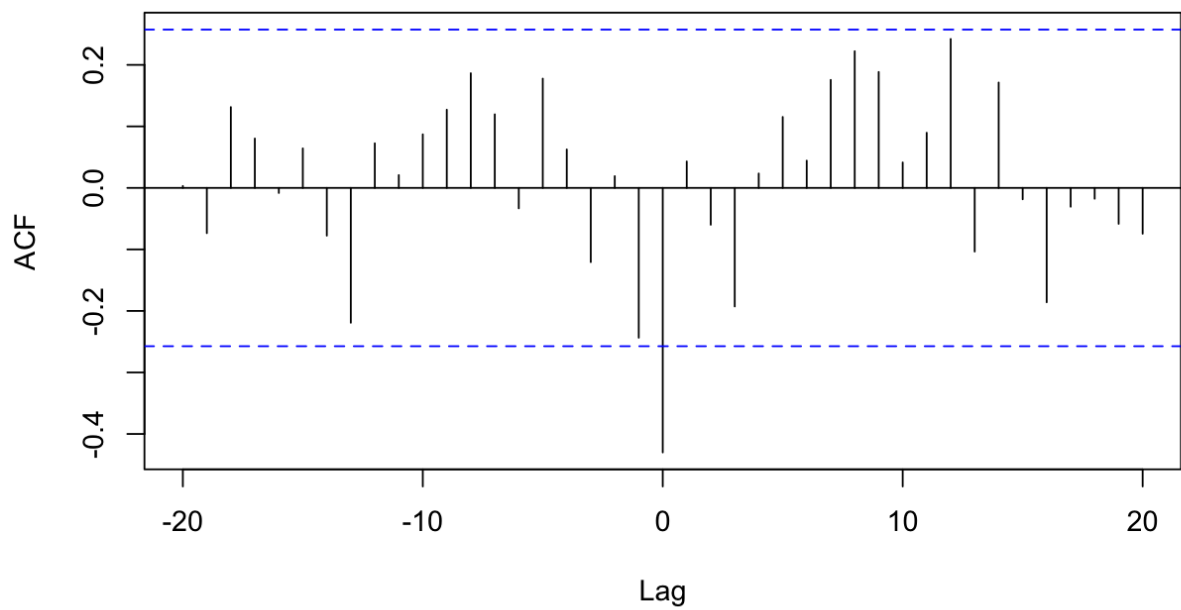


Figure 16

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